

The Laplacian Operator

The Laplacian operator ∇^2 is applied to scalar functions and it actually describes two consecutive operations; The gradient ∇ , which gives the direction in which the function increases, followed by the divergence $\nabla \cdot$ of that gradient. So it describes how much the *increase* of the function *goes out* of a specific point, then if the Laplacian is positive the surrounding points are mostly greater than this specific point, and vice versa.

The Heat Equation

Once the intuition of the Laplacian is clear, then describing the behavior of heat in a medium becomes very simple. any point in the medium is only heated if the surrounding points are hotter on average.

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

We'll expand the Laplacian while only considering the 2-dimensional form of the problem, since we're trying to find the equilibrium state of the system we can set $\frac{\partial u}{\partial t} = 0$, the following equation is called **Laplace's equation** and it can be used to model a heated plate after a long time for example.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Finite Difference Method

We know that the derivative is defined as $\lim_{h \rightarrow 0} \frac{u(x+h,y) - u(x,y)}{h}$, so it can be approximated with a small finite difference as follows:

$$\frac{\partial u}{\partial x} \approx \frac{u(x+h,y) - u(x,y)}{h}$$

By applying this formula again to $\frac{\partial u}{\partial x}$ we can get an approximation of $\frac{\partial^2 u}{\partial x^2}$, you can try making this step yourself before seeing the result.

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u(x+h,y) + u(x-h,y) - 2u(x,y)}{h^2}$$

Similarly:

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{u(x,y+h) + u(x,y-h) - 2u(x,y)}{h^2}$$

Substituting these numerical approximations into Laplace's equation we get this:

$$u(x+h,y) + u(x-h,y) + u(x,y+h) + u(x,y-h) - 4u(x,y) = 0$$

Notice that this equation subtly assumes that all the h 's are equal (the points are placed on a regular grid), the algebra would be more complicated if the differences in the y -direction are not equal to the differences in the x -direction, it would even more complicated if the differences in the same direction (x or y) are inconsistent (the h in the first derivative is different from the one in the second).

You can try the derivation yourself but this is the final general equation, using $u(x,y) = u_{0,0}$ for simplicity.

$$\frac{u_{-1,0}}{h_{x1}(h_{x1} + h_{x2})} + \frac{u_{1,0}}{h_{x2}(h_{x1} + h_{x2})} + \frac{u_{0,-1}}{h_{y1}(h_{y1} + h_{y2})} + \frac{u_{0,1}}{h_{y2}(h_{y1} + h_{y2})} - u_{0,0} \frac{h_{x1}h_{x2} + h_{y1}h_{y2}}{h_{x1}h_{x2}h_{y1}h_{y2}} = 0$$